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The role of environmental statisticians in environmental policy: the case of performance measurement

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ABSTRACT

Environmental statisticians often experience the difficult task to provide clear results for environmental policy objectives on the basis of complex science, limited data, and many sources of uncertainty. This paper highlights the particularly challenging situation of environmental statistics in developing countries before focusing on the role of statisticians working at the nexus of environmental statistics and policy within the context of measuring environmental performance. Three issues create particular tension at the environmental science-policy junction: (1) the general complexity of environmental problems (including scientific uncertainty and insufficient data) and the often negative perceptions associated with their solution, (2) the comparatively recent introduction of quantitative methods and information to the environmental policy process and continued skepticism and ideologically motivated resistance to their routine integration, and (3) the language barriers between policymakers and statisticians as well as between statisticians and other scientists engaged in environmental research. Based on the case study of the 2006 Pilot Environmental Performance Index (EPI) developed by an interdisciplinary team at the universities Yale and Columbia the paper examines how the EPI tackles these tensions while aiming to be a fact-based, statistically sound policy tool that helps countries achieve environmental objectives by tracking progress, identifying environmental "best practices", and providing strategic peer-group analyses. In conclusion, the paper suggests some remedies for the deficiencies in the co-operation between policymakers and environmental statisticians.

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1. Introduction

"What gets measured gets done." "If you cannot measure it, you cannot manage it." Generally associated with situations of economic or corporate interest, phrases like these are nowadays increasingly heard in the environmental policy field. Although the driving forces of the ongoing "measurization" of environmental policy are manifold, including resource constraints, promotion of accountability, and reducing scientific uncertainty, the general trend towards greater use of and reliance on factual data – as well as statisticians and others trained to analyze them – is arguably a positive one.

1.1. *The issues*

Measuring important environmental policy objectives – from climate change to biodiversity, water quality, and ecosystem degradation due to air pollution – tends to be a difficult task. What are the main issues that hamper the use of quantitative data and scientists in the environmental policy process while other disciplines such as economics and the health sciences apply them so effectively to their advantage?

For one there are the twin factors of complexity and perception. Environmental science can be overwhelmingly complex and uncertain. But while this true for economics and

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medicine as well, they enjoy a generally positive image among decisionmakers and the public. The objectives of environmental policy to protect and preserve the natural environment on the other hand often call for behavioral changes that carry negative perceptions because they seem to deny or limit options rather than generate possibilities or cures. Aside from lip-service, actual implementation of solutions to often complex environmental problems is thus only embraced by a relatively small group of environmentalists. As Shellenberger and Nordhaus state in their essay "The Death of Environmentalism": "We believe that the environmental movement's foundational concepts, its method for framing legislative proposals, and its very institutions are outmoded. Today environmentalism is just another special interest" (Shellenberger and Nordhaus, 2004, p. 8).

Second and tied to the negative perception problem is the fact that constructive, data-driven environmental debates are still hampered by ideological and emotional prejudices. As a result, the quantitative and methodologically rigorous study of environmental phenomena arrived comparatively late at the environmental policy table. In contrast the natural sciences have applied systematic, quantitative experiments, and observational studies from the beginning. Most social sciences have followed suit and some of them, such as political science, have by now embraced quantitative methods and scientists among them to such an extent that nearly no theoretical argument is complete without the support of quantitative analyses.

Third, the divide between environmental policy and science has contributed to persistent language barriers between environmental statisticians on one side and policymakers and subject-area scientists on the other. The negative consequences of these barriers include varying understanding of technical concepts, miscommunication, and lack of trust. It is deepened further by the limited influence environmental statisticians have on the framing of policy questions, selection of appropriate study design, data collection and analysis, and the ultimate interpretation of the findings.

In light of the growing environmental challenges such as global climate change, it is therefore timely to investigate the role of environmental statisticians in the creation of environmental information and policies and to identify strategies statisticians can bring to bear on bridging the data-policy gap.

1.2. *Scope and structure of the paper*

This paper focuses on the role of statisticians working at the nexus of environmental statistics and policy within the context of measuring environmental performance. It reviews experiences that an interdisciplinary team at the universities Yale and Columbia has made in measuring country-level environmental performance and translating it into policy messages. Specifically, it examines the recently published Pilot 2006 Environmental Performance Index (EPI), which is aimed to be a fact-based, statistically sound policy tool that helps countries achieve environmental objectives by tracking progress, identifying environmental "best practices", and providing strategic peer-group analyses. The EPI is therefore a suitable case study for the demand on environmental statisticians to create measures that balance methodological transparency and the delivery of clear policy messages to the environmental community, on the one hand, with the use of appropriate statistical techniques and dealing with many sources of uncertainty on the other (Esty et al., 2006).

The paper is organized as follows. Section 2 begins with an examination of the difficulties to define environmental performance in measurable terms and proposes to transfer relevant concepts and methodologies from economics and other social sciences.

Section 3 focuses on environmental policy and its information needs. Since environmental data remain scarce both in quantity and quality, it is useful to focus for a moment on the particularly challenging problems of developing countries to produce and apply reliable statistical data for policymaking.

Despite many persisting problems, the trend towards assessing environmental policy through performance measures continues unabated. Section 4 reviews the Pilot 2006 Environmental Performance Index (Esty et al., 2006) as a potential model for both developed and developing economies interested in tracking progress on environmental objectives. In this context, the need to create more policy-relevant indicators while coping with data gaps and uncertainties is considered as one of the areas of special importance.

The paper concludes with an assessment of the reciprocal relationship between environmental policymakers and statisticians and how it could be strengthened.

2. Definitions and measurement

2.1. *Problems defining environmental performance*

Many different definitions of environmental performance can be found (Commission for Environmental Cooperation, 2000; Global Environmental Management Initiative, 1998; Schulze, 1999; Segnestam, 1999; United Nations Economic Commission for Europe, 2006). General agreement exists that performance refers to the assessment of actual (measured) results of a system against policy, objectives and targets (International Organization for Standardization, 2004). Variations exist with respect to the definition of the system (e.g., country, corporation, household), including its inherent characteristics, boundaries (e.g., closed or open to energy and material flows), and inclusion of effects from exchanges with other systems (e.g., trade and transboundary movement of people, pollution, and capital). For example, while some country-level performance assessments consider only the achievement of domestic environmental objectives, others request that transboundary effects such as long-range air pollution and resource exchanges through trade be taken into account.

The accomplishment of stated objectives or targets in environmental performance measurement refers to the effectiveness of environmental policies. In addition, economic performance assessments evaluate the efficiency with which the objectives or targets have been attained. This is routinely done within the economic theory of maximizing net social welfare, which expresses costs and benefits in monetary terms. Transfer of this approach to environmental issues suffers from a number of limitations, including the lack of

markets for many environmental services, the ''public good'' nature of many environmental goods and services, and the difficulties to objectively estimate indirect as well as long-term costs and benefits[1]. Thus, many environmental performance assessments explicitly or implicitly exclude efficiency criteria from their protocol.

Another aspect relevant to the definition of environmental performance is the intended purpose of the assessment. Compliance monitoring of regulated economic actors has increased in importance over the past decades (Karkkainen, 2001). Other uses include the setting of regulatory standards, identification of ''best practices'', and stimulation of technological innovation (cf. Esty, 2002, for an analysis of the uses of performance measurement in an environmental context). Depending on the purpose of the assessment, measuring performance can range from a simple dichotomous variable indicating success or failure to comply with a regulatory requirement to complex cost/benefit analyses derived from detailed input/output statistics or material flow analysis. As discussed in Section 4 below, the Pilot 2006 Environmental Performance Index falls in between these two categories. It is a data-intensive, non-monetary evaluation of the effectiveness of environmental policies to achieve desirable outcomes in six core policy categories ranging from protecting public health to building a sustainable energy strategy.

2.2. *How does environmental performance measurement differ from economic and social performance assessments?*

The previous section already touched upon the notion of efficiency as an important criterion in performance assessments in economics. Economists estimate efficiency in monetary units to make different policy choices comparable. Although progress is being made[2], efficiency is not yet a standard component of environmental performance assessments. Environmental economists have applied, adapted, and expanded the theory of efficiency to the environmental field. The relatively new disciplines of environmental and natural resource economics have made substantial contributions to the development of new measurement methods, which in turn inspire more efficient environmental policies. Yet, methodological problems for estimating the costs and benefits to society of the impacts of the environmental policy and the large amounts of data necessary to conduct such estimations still constitute significant obstacles to applying efficiency criteria to environmental policy.

Many other approaches to performance assessment can be found in the social sciences. For example, standardized educational aptitude tests, health attainment scores, psychological evaluation protocols for sensitive occupations, and civil liberties indicators are built on extensive and growing bodies of research on the statistical modeling of multi-dimensional concepts. Indicators serving as representations or proxies for latent concepts such as intelligence, and other behavioral traits are ubiquitous and have gained widespread albeit not undisputed acceptance. The exploration and aggregation of environmental and related socio-economic data in the form of indicators and indices has experienced an unprecedented level of attention since the first United Nations Conference und the Human Environment in Stockholm in 1972. And since the majority of environmental policy concerns including climate change, biodiversity loss, land degradation, and water resource management are also multi-dimensional, their description and measurement could benefit from the testing and adaptation of measurement instruments developed in the social sciences at large.

Are there other aspects that models in environmental and the social sciences have in common and can we learn from these linkages? Without laying claim to comprehensiveness and detail, there are a few similarities, which may have potential for the cross-disciplinary transfer of methods. For one, the examples of multi-dimensional concepts cited above also share their intrinsic complexity to measure them. Intelligence tests, for example, vary in their operationalization of the concept of intelligence. Useful insights could emerge from an examination of their design and the historical processes through which these tests have evolved into accepted measurement tools.

Another common element of the environmental and social sciences is that the systems and processes are often not amenable to experimental control and replication. For example, many studies in both fields rely on ex-post observational data and cannot be replicated. Endogenous factors creating 'noise' in the system under investigation cannot easily be reduced and limit comparability across systems. Small sample sizes further hinder the identification of significant differences in the data. Statisticians have developed a rich body of theory and applications relevant to these and other issues. Examples thereof that are particularly useful in the context of analyzing environmental data for decisionmaking are methods for imputation and estimation of incomplete data (Dempster et al., 1977; Hartley and Hocking, 1971; Ibrahim et al., 2005; Little and Rubin, 2002; Rubin, 1996; Schafer and Graham, 2002), variance–covariance estimation in spatial-temporal models (as discussed by Guttorp, 2000), and techniques to optimize sampling designs, increase statistical power, and support economic evaluations of alternative sampling schemes (Benedetti-Cecchi, 2001; Fox, 2001; Smeach and Jernigan, 1977).

Lastly, a common characteristic of performance measurement in the environmental and social science fields is the occurrence of processes over a broad range of time periods. Many ecological processes take decades, centuries or longer to complete while others occur within days or less. Similarly, behavioral studies record events over decades or hours. In both the possibility of spurious correlations or loss of 'signal' due to noise is large. How other disciplines eliminate or reduce such problems could offer solutions to the environmental field.

[1] Cost/benefit assessments of long-term effects such as climate change require the use of discount rates in order to express costs and benefits arising in the future in net present value terms. The discount rate is the interest rate that reflects the ''value of time'', i.e., the increase/decrease in the value of one dollar not spent today (t_0) but at time $t_0 + \Delta$, $\Delta > 0$. Its choice is not trivial and influenced by assumptions about economic and technological change as well as societal values and beliefs.

[2] Cf. for example, US EPA's *ex-ante* and *ex-post* assessments of the efficiency of Title IV of the Clean Air Act, US EPA, http://www.epa.gov/airmarkets/articles/healtheffects/index.html (accessed 24 April 2006).

3. Policymaking and environmental statistics

3.1. *The search for good data and ugly facts*

It is not newsworthy to lament the continued lack of environmental and related socio-economic data. Although dedicated scientists, field observers, and technicians constantly fill gaps in the environmental knowledge puzzle, this lack remains a serious problem, especially with respect to the measurement of cause-effect and dose–response relationships as well as performance measurement. Often, the necessary data are not available because they cannot be measured directly, are too expensive to measure, information is not shared for reason of security interests, or the necessary data have not been collected due to faults in the planning process or scientific unknowns at the time of measurement. Other factors that pose additional burdens on the work of statisticians include measurement error, incompatibility of monitoring data across jurisdictions, inconsistent laboratory analysis techniques, a general lack of policy-relevant indicators, but also a frequently encountered resistance or negligence to incorporate statisticians' input early on in study design and data collection processes.

Environmental statisticians, therefore, often experience the difficult task to provide clear results for policy objectives on the basis of complex science, limited data, and many sources of uncertainty. The problem is far from new: Arnold Zellner in a 1992 editorial in JASA calls on statisticians to renew their attention to ''get the facts'' in the form of reliable data (Zellner, 1992). This should not be equated with a call to eradicate all sources of errors in the data (it would be prohibitively expensive and infeasible) but that such errors are acknowledged in the subsequent analysis, at least in the discussion of the findings but preferably through estimates of standard errors and confidence intervals.

Furthermore, statistical data exploration and analysis comprises not only the search for neat patterns consistent with the investigators' personal opinions or even with society's broadly shared beliefs. Quoting Zellner (1992), environmental statisticians should strive to produce ''ugly facts'' and find explanations for them. After all, a new index of environmental performance that only confirms existing knowledge does not provide any new or additional value. Zellner advocates that we test existing theory, hypotheses, and beliefs more rigorously and develop new explanations for them if we find patterns inconsistent with our existing knowledge. Because ugly facts are easily dismissed as outliers, measurement errors, or other nuisances, they are often eliminated from the analysis. Distinguishing ''good'' from ''bad'' data is an exploratory process that requires the close interaction of subject-area experts, statisticians, and policymakers. In fact, path-breaking new insights into environmental processes have come about through the collection of good data and their analysis in interdisciplinary teams. An example is the discovery of the ozone hole over the Antarctic by British scientists Farman, Gardiner, and Shanklin (The Ozone Hole Inc., 2006). The drop in ozone concentrations in the 1980s was initially attributed to faulty measurement instruments. Instead, thorough (statistical) analysis revealed a seasonal and widening area of thinning stratospheric ozone concentrations, which prompted a rare case of global environmental co-operation to gradually phase-out the production of six major CFCs[3].

As the above example demonstrates, environmental policy – perhaps more than other policy areas – is concerned not with average but with extreme events. This requires specialized analytical tools and statisticians are pre-disposed to contribute to their development. Zidek (2006), for instance, promotes the term post-normal statistical science (PNS) for the new era of increasing intertwining of environmental statistics and policy. PNS refers to the complexity of environmental policy questions, the high level of data and scientific uncertainty surrounding them, and the presence of a plurality of divergent societal values and policy perspectives. As he puts it succinctly, it ''. . . embraces issue-driven scientific inquiry related to environmental controversies; the facts are uncertain, values in dispute, stakes high, and decisions urgent'' (Zidek, 2006).

Environmental performance measurement exhibits the typical symptoms of PNS and statisticians are, therefore, needed even more urgently to contribute their knowledge of analyzing extreme values, dealing with uncertainty, and testing model sensitivity to the decisionmaking process that is needed to arrive at meaningful, consensual policies. Those working on the design of environmental indicators for performance or sustainability are indeed breaking new ground in ''mandated science'', i.e., science at the intersection of values, policy, and traditional, independent core science (Zidek, 2006). Mandated science inevitably requires statisticians to balance scientific impartiality with the scope and purpose of the research they are carrying out. Frameworks for environmental performance assessment, for example, are influenced by the worldviews of those developing them and/or possibly those funding their development.

Emphasizing the collection of good data and an appreciation of unusual, ugly facts might help increase the credibility of environmental performance assessments. This is especially needed given the continued interest but growing skepticism directed at their results. The latter is a direct consequence of the growing number of indicator sets for measuring sustainability at large that Pinter et al. (2005) call the ''indicator zoo'' and that are developed from limited information, different frameworks, and whose interpretation subsequently generates mixed messages about environmental conditions and likely future trends (Wilson et al., 2007).

3.2. *Environmental statistics and environmental policy from the perspective of developing countries*

Chapter 40 of Agenda 21 (United Nations, 1992) put the need for environmental information center stage in the international environmental and development theater. In particular, world leaders at the 1992 United Nations Conference on Environment and Development (UNCED) in Rio de Janeiro, Brazil, acknowledged that the gap in data availability, quality, and accessibility is widening between developed and developing countries. There is a growing body of research into the

[3] Refer to http://ozone.unep.org/Treaties_and_Ratification/2B_montreal_protocol.asp for more information on the science and policy process surrounding the Montreal Protocol on Substances that Deplete the Ozone Layer.

drivers and implications of the knowledge divide between the global North and South (cf. Karlsson, 2002 for an overview on this subject).

To counter the information gap, a number of international action plans were developed, data collection programs implemented, and environmental ministries and agencies created but 15 years later, the gap is not closing. Environmental performance measurement today is thus confronted with a seeming abundance of data for industrialized or developed countries (seeming because the available data are not always the most needed) and a substantial information scarcity for the rest of the world[4]. In the following, I therefore review specifically the situation of environmental statistics in the developing world.

In the aftermath of the Stockholm and Rio conferences, a general ''stock-taking'' of available environmental information took place in many countries; often under the leadership of the national statistical office. Both OECD and developing nations identified the existence of considerable cross-sectoral dispersion of environmental information. This fragmentation of data collection efforts across line ministries and governmental agencies (agriculture, forestry and fisheries, health, planning and development, tourism, etc.) is both helpful and problematic as it means that a variety of data pertaining to environmental issues already existed but was generally not collected specifically for environmental policy purposes. Indeed, much information to this day is stored in incompatible formats and databases, access-restricted, and organized according to incompatible classifications or definitions. An example of the dispersion of environmental information is the measurement of pollution as one of the first environmental policy objectives (Polfeldt, 2006). Yet, data on pollutant emissions, their ambient concentrations, and ultimately their health effects are scattered across different governmental agencies, collected according to varying protocols, or missing. A frequently observed upstream problem of information dispersion is a general lack of coordination across sectors and ministries, which leads to the duplication of efforts and critical information gaps elsewhere.

Nevertheless, the earlier and more successful growth of environmental movements in OECD countries, including the founding of ''green'' political parties, sparked consolidation efforts sooner and at a faster pace compared to developing countries[5]. Another factor that limited the effective production and use of environmental information was the relative weakness of national statistical services in many developing states, particularly with respect to the necessary expertise in environmental data collection and analysis.

Polfeldt (2006) points out that regardless of the many obstacles, at least 30 developing countries released State of the Environment (SoE) reports in the aftermath of the Rio Summit. Many of those, however, remained one-time efforts because the countries could or would not consider sustaining them in the long-term.

International, regional, and bilateral capacity building projects over the past two decades have generated multiple benefits for environmental statistics as a new field of official statistics in developing countries. They helped integrate environmental concerns into national policy strategies; built legitimacy for newly created environmental ministries and environmental protection agencies, improved cross-sectoral communication channels, and resulted in the training of critically needed environmental statisticians[6]. More substantial environmental reports with direct links to policy agendas are now published, albeit data quantity and quality still lacks OECD standards. They still often contain more *pressure* data such as population size, industrial activity, and extent of trade, which are generally easier to find than *impact* and *outcome* figures such as pollutant concentrations in air and water.

I believe it is fair to assume that environmental statisticians in developing countries face similar but also additional challenges compared to their colleagues in the OECD member states. In the latter, people often express environmental concerns as among their highest priorities for policy action and environmental polls regularly find a widespread appreciation of environmental values (e.g., the environmental poll conducted by the Yale Center for Environmental Law & Policy, 2005). This creates public pressure on political actors to invest resources on issues such as environmental monitoring, data collection and analysis systems, training, and general environmental infrastructure building. In contrast, mainstreaming environmental concerns into policymaking is an uphill battle for many developing nations struggling to meet basic human needs. The environmental indicators available in these countries are, therefore, often only those mandated and/or funded through international or bilateral training and assistance programs or are part of international sustainable development strategies such as the Millennium Development Goals (United Nations, 2000). It is noteworthy that these indicators have been developed with global comparability in mind, to prod countries into action, or to facilitate the direction and allocation of development aid rather than their utility for individual countries. It remains to be seen if such measurement efforts create sustained benefits to the countries as well as a sense of national ownership or if they place undue burden on already overloaded statistical services (Statistical Commission, 2002).

[4] Cf., for example, the extensive series on national environmental country reviews published by the Organization for Economic Co-operation and Development (OECD) online at http://www.oecd.org/department/0,2688,en_2649_34307_1_1_1_1_1,00.html (accessed 27 April 2006).

[5] In developed countries the foundation for much of the currently existing environmental law and regulatory systems was created in the late 1960s and throughout the 1970s. In contrast, environmental law and regulation in developing nations and emerging economies is a recent phenomenon originating in part in increasingly destructive natural resource exploitation, rising pollution levels, and growing environmental and civil society movements that are sometimes supported by a general democratization of the country (cf. United Nations Development Program, 2002. Human Development Report 2002—Deepening democracy in a fragmented world. New York, UNDP.

[6] ... Although the author's personal experience and anecdotal evidence indicates that the training of environmental statisticians alone is not sufficient for building national environmental capacity because trained personnel often leaves for better-paying jobs elsewhere.

Table 1 – The EPI's six policy objectives, their constituent indicators and corresponding performance targets

Policy objective	Indicators contained in policy objective	Performance target for the indicator
Environmental health	• Indoor air pollution	0% of households
	• Access to drinking water	100%
	• Adequate sanitation	100%
	• Child mortality	0 deaths/1000 children 1–4years
	• Particulates concentration	10 $\mu g/m^3$
Air quality	• Particulates concentration	10 $\mu g/m^3$
	• Regional ozone concentration	15 ppb
Water resources	• Nitrogen loading	1 mg/l
	• Water consumption	0% oversubscription
Biodiversity and habitat protection	• Water consumption	0% oversubscription
	• Wilderness protection	90% of wild areas protected
	• Eco-region protection	10% for all biomes
	• Timber harvest rate	3%
Natural resources productivity	• Timber harvest rate	3%
	• Agricultural subsidies	0%
	• Overfishing	Score of 0 (no overfishing)
Sustainable energy	• Energy efficiency	1650 TJ/million US$ GDP
	• Renewable energy	100%
	• CO_2 emissions	0 net emissions

Note: The indicators particulates concentration, water consumption, and timber harvest are each constituent to two policy objectives. They and their corresponding performance target are therefore listed twice. The total number of indicators in the EPI is 16.

Even with limited resources and data, environmental statisticians in developing countries need to be trained in environmental policy or paired with policymakers to better link available information with policy goals. In the following section I present our experiences in the development of the Pilot 2006 Environmental Performance Index as an illustration of the challenges and opportunities arising in linking environmental policymaking to data-driven performance measurement.

4. Lessons from the 2006 Pilot EPI

The Pilot 2006 Environmental Performance Index is produced by the Yale Center for Environmental Law and Policy and Columbia University's Center for International Earth Science Information Network (CIESIN)[7] as a tool to assist policymakers in tracking their environmental performance.

The EPI is a global index in the sense that environmental performance is evaluated for all 133 countries included in the index using a uniform set of six widely established environmental policy objectives: environmental health, clean air, clean and sufficient water, protected biodiversity and habitat, productive natural resources, and sustainable energy. Furthermore, within each of the policy categories between two and five indicators track progress against fixed performance targets (summarized in Table 1).

The choice of the policy categories is informed by international policy consensus (e.g., the United Nations Millennium Development Goals) as well as the developers' belief that these are areas of environmental policy that every country should strive to make progress in.

The EPI indicators were chosen to as much as possible measure on-the-ground conditions since outcome measures are the ultimate gauge of successful policy (Esty et al., 2006). Performance targets are specified for each indicator on the basis of international agreements, the scientific literature, and expert consultation to promote long-term desirable public health and ecosystem sustainability.

Performance analysis becomes more meaningful when metrics can be compared at different levels of aggregation, across sectors, and against different reference units. This requires that the indicators are comparable, e.g., by measuring them on the same scale. For this reason, a two-step procedure converts the EPI indicators to proximity-to-target values, which reflect how closely current performance approaches the targets. First, for each indicator country values below the 5th percentile were set to the 5th percentile value to avoid outliers skewing the distribution and in the few cases where a country exceeded the target, its value was set to the target, thereby not rewarding "overperformance". In the second step, the indicators were transformed to a scale from 0 to 100 using a simple arithmetic procedure such that a final indicator score of 100 is equal to meeting the target and a value of 0 corresponds to the worst possible or worst observed value[8].

The EPI allows performance analysis at three levels of aggregation. At the first step, scores for the six policy categories are calculated as the weighted average of the corresponding proximity-to-target indicators. The weighting applied in this step is derived from both expert consultation

[7] The EPI was further supported by the World Economic Forum, Switzerland, and the Joint Research Centre of the European Commission, Italy. The full report is available at http://www.yale.edu/epi/ (last accessed January 29, 2007).

[8] If the range of the indicator is a known fixed interval, then this interval is used in the calculation of the proximity-to-target value. Otherwise, the range between target and worst observed value is used.

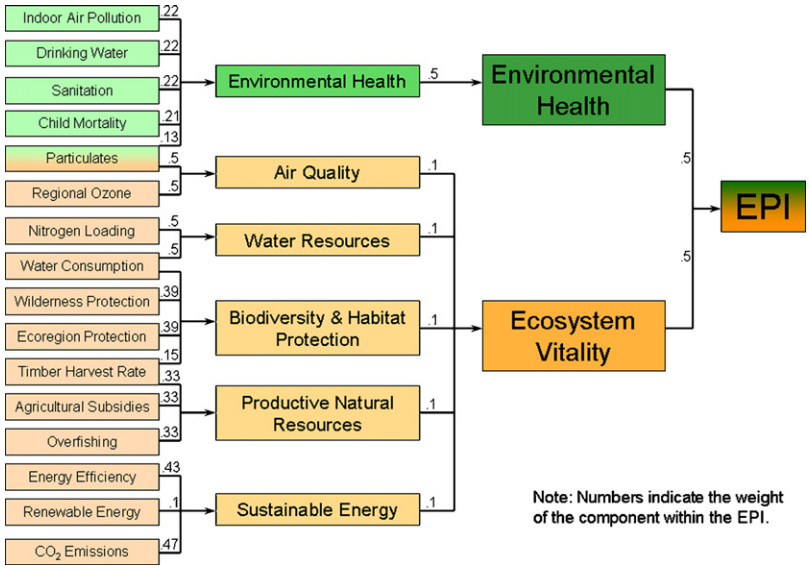


Fig. 1 – Constructing the Pilot 2006 Environmental Performance Index (EPI).

and statistical analysis using Principal Component Analysis (PCA). The policy categories are then combined to two broad-based objectives, i.e., ensuring environmental health and ecosystem vitality. Finally, the policy scores are averaged to yield the Environmental Performance Index. Fig. 1 shows a schematic representation of the EPI framework and Table 2 displays the resulting country ranking in reduced form.

The EPI's utility as a policy assessment tool was paramount in its development process. Although it is a global index similar to the MDG's, individual use at country level is maintained and even recommended, for example, by reflecting national priorities through different weighting of the EPI's components. At the same time, cross-country comparability is ensured through the choice of uniform performance targets.

All indicators have undergone a rigorous selection and development process that included policy relevance and data availability and quality. Most data are drawn from internationally recognized sources or custom-built by a network of experts consulted in the production process. Thereby, reporting burden on developing countries has not been increased. Nevertheless, the Pilot EPI represents work in progress and substantial challenges remain. In the following sections I review the status of environmental performance measurement and the lessons learned from the EPI with a view toward describing the role of statistics and statisticians in advancing the field.

4.1. *Coping with data gaps, lack of quality, and relevance*

Any performance assessment can only be as good as the data that feed into it. The three most pressing data shortcomings affecting the EPI are:

- Lack of policy-relevant indicators (or the data for them).
- Incomplete data sets, i.e., with respect to temporal and/or spatial coverage.
- Varying levels of data quality, specifically as it relates to measurement error and cross-country comparability of data vis-à-vis underlying definitions, data collection methodology, and quality control protocols.

Since environmental statistics is a relatively new discipline (Guttorp, 2000), it means that time series of environmental indicators are often short or intermittent. The cost of environmental monitoring activities and their considerable technical requirements also contribute to the inadequacy of the quantity of available information especially in developing countries (cf. Section 3.2 on the disparities in environmental data availability in developed and developing countries).

Harmonization efforts and data quality standards also have had a fairly short history in environmental statistics compared to other disciplines. Although regional and international organizations such as the Statistical Office of the European Communities (EUROSTAT) and the United Nations Statistics Division (United Nations, 2004) are actively promoting their development and implementation, comparability of environmental data continue to be very limited.

Lastly and perhaps most importantly, despite investment and successes in building environmental information systems and databases (cf. Section 3.2), policy-relevant indicators are scarce. Among the reasons for the disconnect between data collection activities and policymaking are

- Policy-specific indicators tend to be more complex, their scientific basis is more difficult, and the necessary information harder to collect.
- The channels of communication between subject-area experts, statisticians, and policymakers are insufficient and not reciprocal in the sense that policymakers seek statistical advice throughout all stages of the policy development process and statisticians actively see to contribute to apply statistical theory and applications to the policy process.

To address these issues, the development procedure of the EPI involved broad consultation with experts and policy-

Table 2 – Selected EPI countries ranked by EPI score

Rank	Country	EPI score	Peer group	Environment health	Air quality	Water resources	Biodiversity and habitat	Productive natural resources	Sustainable energy
1	New Zealand	88.0	OECD	97.9	83.7	98.8	73.4	61.4	73.4
3	Finland	87.0	OECD, EU	98.8	65.3	99.5	54.3	81.5	75.7
5	United Kingdom	85.6	OECD, EU	98.9	61.6	91.9	58.8	71.6	77.8
8	Canada	84.0	OECD, FTAA	98.6	56.2	98.4	55.1	73.9	62.8
14	Japan	81.9	OECD, POP, ASEAN	97.6	52.6	94.8	70.4	33.3	79.7
15	Costa Rica	81.6	FTAA	81.1	60.6	100	80.2	83.1	86.0
26	Chile	78.9	FTAA	87.2	63.7	83.7	68.3	63.0	74.6
28	United States	78.5	OECD, FTAA	98.3	44.7	73.9	66.8	38.9	69.7
33	Hungary	77.0	OECD, EU	94.2	55.6	77.0	47.6	50.0	69.2
34	Brazil	77.0	FTAA	79.3	64.0	97.8	50.4	80.9	80.6
36	Lebanon	76.7	POP	93.4	52.1	89.3	20.2	76.6	61.2
46	Gabon	73.2	AU	61.0	96.1	99.9	62.5	88.9	79.8
51	Ukraine	71.2	NIS	93.8	56.6	65.2	40.0	77.8	3.7
53	Iran	70.0	D	85.7	31.1	72.4	47.9	83.3	36.6
64	Jordan	66.0	D	85.5	40.6	45.8	55.9	38.0	51.7
72	Ghana	63.1	AU	48.8	87.3	99.4	50.1	67.5	83.3
78	Uganda	60.8	LDC, AU	31.7	98.0	92.7	73.6	93.0	92.4
80	Kyrgyzstan	60.5	NIS	53.7	50.6	92.7	68.0	100	38.3
81	Nepal	60.2	LDC, POP	44.1	35.9	99.0	60.5	99.0	86.4
85	Egypt	57.9	D, AU	74.6	14.8	71.5	23.9	38.9	57.2
89	Rwanda	57.0	LDC, POP, AU	31.1	91.1	95.0	63.2	77.3	87.3
94	China	56.2	ASEAN	61.0	22.3	49.6	68.1	66.2	50.8
118	India	47.7	POP	43.8	28.4	67.6	39.7	62.1	59.7
127	Pakistan	41.1	D	46.1	8.2	37.9	23.0	44.4	66.6
133	Niger	25.7	LDC, D, AU	1.0	22.9	56.6	38.9	50.0	83.6

Shown are also the scores for the six policy objectives: environmental health, air quality, water resources, biodiversity and habitat protection, productive natural resources, and sustainable energy. *Note*: to place the EPI and policy category scores into appropriate context selected peer group memberships are provided in the 4th column. They are in order of appearance: OECD, organization for economic co-operation and development; EU, European union; FTAA, free trade agreement of the Americas; POP, high population density; ASEAN, association of Southeast Asian Nations plus China, Japan, and South Korea; AU, African Union; NIS, Russia and newly independent States; D, desert countries; and LDC, least developed countries.

makers from the outset. Virtual expert networks were created focusing on the policy categories that worked collectively to identify the best currently possible indicators within their domain. The result is not only an improved indicator set but also a ''wish-list'' of relevant but as of yet unavailable sectoral measures of environmental performance that is used as a blue print in our ongoing efforts to fill critical data gaps.

The EPI team also investigated the imputation of missing values to increase geographic coverage. Despite the existence of a theoretical foundation for missing data imputation and a growing availability of models accommodating different missingness patterns (Demirtas, 2004; Ibrahim et al., 2005; Rubin, 1996; Schafer, 1997), there remain a number of modeling challenges. In addition, political confidence in this type of modeled data is still limited. The pilot EPI thus relies solely on observational data while the use of imputation models is further studied for future releases.

4.2. *Reconciling transparency and simplicity with statistical rigor*

The use of statistical methods in index development to answer environmental science and policy questions is uneven across and within disciplines. Some applications worth mentioning are, inter alia, in ecology, the soil and water sciences, as well as in biodiversity protection. For example, based on a review of the statistical properties of widely used indices measuring species richness and abundance Buckland et al. (2005) develop a set of criteria an index should meet and show that a modified version of the Shannon index (Shannon and Weaver, 1949) has these properties. Chiu and Guttorp (2004a,b) build on the Benthic Index of Biotic Integrity (B-IBI) by Kerans and Karr (1994) and develop a stream health index with reduced bias, variability, and reduced sensitivity to changes in the metric scoring system used to construct the index. Soil scientists have developed indices for soil biota stress, droughts, and land management (Hortensius and Nortcliff, 1991; Orwin and Wardle, 2004; Smakhtin and Hughes, 2004; Syers et al., 1995). Washington (1984) reviews several performance indices pertaining to aquatic ecosystems.

Other areas of environmental relevance, specifically sustainable development, and within its realm the theme of environmental sustainability and performance measurement, have produced a rapidly growing number of indicators and indices but while policy relevance was a primary focus, few of these efforts considered statistical properties such as bias, variance, and robustness to changes in the underlying assumptions explicitly in the design process (Parris and Kates, 2003; Pinter, 1994; Pinter et al., 2005; UNEP/RIVM, 1994; United Nations, 1991, 1994, 1996; United Nations Commission for Sustainable Development, 2001).

The credibility of a performance measure hinges not the least on its methodological soundness. While the users of performance indices such as the EPI may not be interested in the finest detail of their calculation, transparency in the methods applied, the data transformations carried out, and the underlying model assumptions is essential for the index to gain credibility and acceptance. There might also exist opportunities for developers of environmental performance indices (and other aggregate environmental policy-relevant measures) to adopt statistical methods from scientific indices in ecology, hydrology, etc. to address aspects such as metric selection, standardization, aggregation, and weighting.

Often, however, transparency is at least partially equated with methodological simplicity, thereby calling for trade-offs between statistical sophistication and ease of understanding by non-technical audiences. The question thus arises whether the model should be built according to the best available knowledge or if it should prioritize simplicity and easy comprehension[9]? The answer is that the appropriate choice of statistical methods depends on the specific circumstances of the project. Indeed, a review of a more broadly defined set of measures used in environmental and development policy suggests that in practice, the solution lies somewhere in between[10]. Interestingly, very few of the papers and sources reviewed were (co-)authored by statisticians, which points to an opportunity for greater involvement of statistical experts in the development process.

Notwithstanding the degree of statistical sophistication, existing performance measurement initiatives make use of a variety of statistical techniques with differing demands on the user's technical knowledge. The EPI, for example, is constructed on the basis of expert opinions and statistical analyses (Esty et al., 2006). The choice of this middle-path is deliberate and reflects our explicit objective to create an easily comprehendible and usable policy tool, which is informed by and reliant on statistical techniques where appropriate and necessary to fulfill this goal. The development process was therefore designed to match broad user buy-in with statistical rigor.

Nevertheless, the EPI is a work in progress, which includes the possibility for greater use of statistical tools once the available data have improved to the extent judged necessary by experts. Potential areas of statistical improvement include:

- Build models for missing data imputation.
- Estimate confidence intervals for index scores taking into account measurement error.
- Test the usefulness of alternative weighting procedures such as Multi-Criteria-Analysis (MCA).
- Expand the scope of sensitivity and uncertainty analysis to include additional factors of sensitivity and uncertainty.
- Develop causal models for the driving forces of environmental performance and other policy interests.

At this point, our general experience suggests that practically useful environmental performance measures that resonate with diverse audiences should be transparent in the data and underlying assumptions and as methodologically simple as possible. Although Zellner's ''keep it sophisticatedly simple'' variation (Zellner, 2007) on the well-known KISS

[9] In this context, also refer to the use of imputation models in Section 4.1. To read more about the risks of imputation, cf. Allison, 1999. Multiple Imputation for Missing Data: A Cautionary Tale. Philadelphia, University of Pennsylvania.

[10] This statement is based on a review of selected literature and electronic sources, which are listed in Appendix I.

statement might offer a guidepost, the EPI sacrificed some opportunities for statistical modeling in favor of a simpler framework that uses the best available indicators.

4.3. *Understanding the role of societal values in measuring environmental performance*

The lack of consensus on how to measure environmental performance (and the broader concept of environmental sustainability) is reflected in the diversity of perspectives and societal values motivating or at least influencing existing definitions (Adriaanse, 1993; Azzone and Raffaella, 1994; Bandura, 2005; Färe et al., 2004; Freudenberg, 2003; Grafton and Knowles, 2004; Hajer, 1992; Hammond et al., 1995; Organization for Economic Co-operation and Development, 1993, 1996; Segnestam, 1999; Tyteca, 1996; World Bank, 1996). As Levett (1998) in his discussion of indicators of quality of life points out "the search for better indicators is not only a technical problem. It is also ... about ... what it is that we value." The role of personal and societal values in environmental policy is also of importance vis-à-vis the concept of post-normal statistical science in that the indicator development process requires not only creative and technical expertise but also room for testing, critiquing, and finding consensus. Hence, measures of environmental performance developed at and for certain times and locales may not be found useful or even agreeable by other groups, in other times, and/or places. The Pilot EPI makes the influence of value judgments and need for further debate explicit (Esty et al., 2006):

> "Derived from a careful review of the environmental literature, these twin goals [environmental health and ecosystem vitality] mirror the priorities expressed by policymakers ... The Pilot 2006 EPI represents a *work in progress* meant to stimulate debate on appropriate metrics and methodologies for tracking environmental performance, enable analysis of the determinants of environmental success, and highlight the need for increased investment in environmental indicators and data."

A number of conclusions can be drawn from the influence of value judgments on the development of measures of environmental performance:

- Performance measures should be reviewed periodically to ensure they represent the goals and values of those they are intended to assist in decisionmaking and whose well-being is being affected by those decisions.
- There is no "right" measure or indicator set. Instead of searching for the universally accepted measure, focus should concentrate to make the measure "fit for purpose" (Levett, 1998).
- Composite measures, which involve weighting of the constituent components should be tested for the effect of the selected weights on the outcome and retain a degree of flexibility regarding the choice of the weights to accommodate different priorities. Such sensitivity analysis should extend to all subjective assumptions made in the design process.

4.4. *Communicating results*

Environmental performance assessments serve a diverse group of users, including scientists, technicians, policymakers, the media, and the general public. This puts different demands on the way the results are prepared and disseminated. Small, homogeneous groups such as scientific communities concerned, for example, with biodiversity protection, use their own technical jargon, graphical presentations, and other tools to effectively communicate with each other. In contrast, performance measures designed to appeal to policymakers and/or the non-trained face substantial diversity in backgrounds, access to communication channels (e.g., Internet, libraries, etc.), interests, and uses.

Too little consideration is sometimes given to they way measurement results are presented and disseminated. Criteria commonly associated with a 'good indicator' often include a communication aspect (Chess et al., 2005). The challenge for the statistician is thus to combine methodological robustness with ease of comprehension. If the assessment is to promote public participation, the issue of competence should be considered at the outset: an indicator is not merely a representation of a phenomenon but also a communication tool in its own right. As Chess et al. (2005) summarizes: "indicators are meant to be representations of knowledge, selected through best available procedures, and easily interpreted. As such, they have the potential to promote 'competence' in public participation." This goal is reflected in a distinct school of political thought. Advocates of *deliberative democracy* promote access to information as a means to strengthen political participation by various stakeholder groups in a bottom-up fashion (Weitzman et al., 2006). In this context, access to information is dependent not only on where the information is stored and through what channels it is disseminated but also on how it is derived at in the first place.

In addition to the methodological framework, new developments in graphical methods and data visualization can be powerful aides in communicating environmental performance assessments. But frequently, the effectiveness of graphical and visual design tools is diminished due to lack of time (i.e., most time is spent on the development of the performance framework, gathering and analyzing the data, and writing the report), expertise (e.g., graphical designers are not often consulted), and considerations of reproducibility and data dissemination (specialized software may not be available to all users of the assessment).

In the case of the EPI, presentation and dissemination were given substantial attention and are considered to have contributed noticeably to its successful dissemination. The EPI consists of a full report and a short policy-focused summary brochure, which is accompanied by a CD-Rom. Both the report and brochure are disseminated in print and as downloadable files through the Internet. To further increase transparency and encourage debate the EPI data are also available by request and for free download.

How widely the final product penetrates targeted user segments also depends on the language used. The EPI avoids technical jargon where it does not contribute to better understanding. For global assessments we found that the language of publication also matters. Although the EPI is

currently available in English only, we are studying the feasibility to translate future releases into other languages to expand into regions where English is not widely spoken.

5. Conclusions

This paper addresses some of the challenges that environmental statisticians face at the nexus of environmental statistics and policy, specifically within the context of environmental performance measurement for decisionmaking. Beginning with a lack of a commonly accepted definition of environmental performance, measurement efforts are further hindered by a number of technical and non-technical problems. The Pilot 2006 Environmental Performance Index is used as an example of a global environmental performance measure experiencing many of these problems.

The dominant concern today remains the need for more policy-relevant environmental indicators of sufficient quality and coverage. Many developing countries, in particular, find themselves unable to fill the underlying data and capacity gaps without assistance but international environmental indicator initiatives run the risk of favoring global comparability over local utility. The resulting reporting requirements pose an additional burden on national statistical services while the sustained benefits of international measurement initiatives is not always certain. Statisticians tasked with developing environmental indicators increasingly require not only an understanding of the underlying science but also the targeted policy objectives in order to create the best measures from available data.

Furthermore, application of sophisticated statistical models without consideration of issues such as transparency, level of existing expertise on the part of the intended users, and aspects of communication and dissemination reduces the potential value of performance assessments. These are non-trivial aspects in that they require public participation and expert consultation and careful balancing of stakeholder preferences. Whenever the measurement objective is not uniformly accepted, the science uncertain, and the need for policy action large, the statistician moves into the area of post-normal statistical science. Being clear about the purposes of the assessment, exact framing of the system to be evaluated, and testing the assessment's robustness against deviations from the framework's assumptions is crucial to build credibility and reduce the growing number of seemingly contradicting measures. In addition, societal values, individual worldviews, and political circumstances influence the process of consensus finding and the outcome should hence be reviewed periodically to reflect changing views and conditions.

5.1. *The role of environmental statisticians in environmental policy*

Environmental statisticians can and should play a dual role in policymaking by supporting and inspiring decision processes. As experts on the collection, analysis, and interpretation of statistical information they generate and analyze the information on which policy choices and regulatory frameworks are built. They further support decisionmakers by providing scientifically derived testimony on the potential risks, benefits, and other probabilistic events and make projections in time and space that assist policymakers in weighing alternatives, balancing competing objectives, and planning ahead. At the same time, advances in statistical theory and methods facilitate environmental research, reduce uncertainty, and create new paths to knowledge discovery. These contributions by statisticians in turn help inspiring new policy initiatives or adjusting existing ones.

Indeed, environmental statistics as a relatively new discipline has made substantial contributions to our understanding of environmental and ecological processes. At the same time the growing complexity and recognition of environmental problems has increased the demands for timely and accurate information at nearly all levels and branches of environmental policy.

So what can environmental statisticians do to address the three issues – negative perceptions of environmentalism as a special interest, resistance to the involve quantitative data, analysis, and statisticians in the policy process, and language barriers – described in Section 1.1 and gain the acceptance that other policy-informing disciplines including economics and the health sciences enjoy?

Environmental statisticians can probably do little to eliminate the sidelining of environmentalism as a special interest. Developing positive visions, values, and solutions to environmental problems that spark enthusiasm and unite people behind common goals is the challenging task of environmental leaders in government, the non-profit and private sectors as well as academia but statisticians can contribute by providing quantitative benchmarks, scenario analyses, and risk assessments. The argument can thus be made that more environmental statisticians are needed in governmental agencies, large corporations, non-profit research institutions, think tanks, and civil society organizations. Such change cannot come overnight and it has to come primarily from a culture-change within these institutions that recognizes the value of involving quantitative scientists in policymaking.

The second issue states that there is a resistance to routinely include quantitative and methodologically rigorous studies in the policymaking process. This can be changed – and is changing – in a number of ways. One possibility could be to build-up demand for environmental statisticians through regulation similar to that found in the medical, health, and pharmaceutical sciences. If, for example, approval of activities that can be presumed to have environmental impacts is tied to quantitative impact assessments investigating and estimating such impacts, the role of environmental statistics and statisticians would be greatly enhanced. This procedural practice of tying policy decisions to statistical analysis is already practiced in several countries and sectors[11]. In addition, applied environmental statisticians could seek to expand the audiences for their work by attending and

[11] See, for example, the United States National Environmental Policy Act (NEPA) of 1969, Sec. 102 [42 USC § 4332]. Available online at http://www.nepa.gov/nepa/regs/nepa/nepaeqia.htm (last accessed January 29, 2007).

presenting at policy-oriented conferences. Knowledge transfer is rapidly evolving and environmental statisticians could harness its tools to present complex, quantitative science in more accessible, policy-relevant ways. This connects directly to the third issue.

Lowering the language barrier and increasing the influence environmental statisticians exert on policy also requires change from both sides. It is necessary for policymakers to change the tendency to ask for too little statistical advice too late in the process. With the expectation to receive unambiguous answers, disappointment arises from the perceived failure of statisticians to deliver such clear guidance based on insufficient experimental designs and data, which then erodes the relevance policymakers give to environmental statistics as a critical source of information throughout the policymaking process. One step to improve the acceptance of environmental statistics is to increase awareness among policymakers that statisticians should participate early on in the data collection, analysis, and interpretation process and that although meticulous planning and analysis can reduce many unknowns, most policy questions remain prone to some level of uncertainty and sensitivity to assumptions.

The barrier is further supported by the view many statisticians hold that the discipline should remain a pure science, unaffected by policy choices, stakeholder views, and political pressure groups. There are many good reasons to limit these influences on the advancement of environmental statistics, considering that the environmental field in particular is a political battleground marred by ideological trenches, fierce debates, and hard-to-argue-against moral attitudes that most statisticians prefer to avoid being drawn into. However, as a consequence, environmental statisticians have not exhausted their role as providers and stimulators of more rational, data-driven policy decisions and have thus foregone the possibility to rationalize policy debates. In order to remain a relevant science in the future and as the stakes to solve environmental problems are growing higher, environmental statisticians are likely faced with the need to become more involved, to understand diverse policy objectives, and to develop the skills needed for communicating statistical material and developing consensus within the context of multiple, competing interest groups, value-laden science, and considerable uncertainties. This may, inter alia, require a review of the curricula of statistics graduate programs but also an expansion of statistical consulting as well as strengthening the interaction with other environmental policy-relevant disciplines, for example, by organizing meetings between statisticians and policymakers at relevant conventions or other suitable venues.

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